

CSE 150A-250A AI: Probabilistic Models

Lecture 6

Fall 2025

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Slides adapted from previous versions of the course (Prof. Lawrence, Prof. Alvarado, Prof Berg-Kirkpatrick)

Review

Approximate inference in loopy BNs

Rejection sampling

Likelihood weighting

Test 1 → Next week
(Week 4)

Scope :
everything till

lec 7 (Thu, Oct 16)

Review

Next week will be
interesting

- test on what we have seen so far
- covering new material
- working on HW .

- Problem

Given a set E of evidence nodes, and a set Q of query nodes, how to compute the posterior distribution $P(Q|E)$?

- More precisely

How to express $P(Q|E)$ in terms of the CPTs $P(X_i|pa(X_i))$ of the BN, which are assumed to be given?

- Tools at our disposal

Bayes rule

marginalization

product rule

marginal independence

conditional independence

Exact Inference: Variable Elimination

- **Idea:** Eliminate **redundant** calculations by storing intermediate results in “factors”.
- A **factor** is a function that takes in values of random variables, and produces a number.
- VE works by eliminating all variables in turn until there is a factor with only the query variable.
- To eliminate a variable:
 - *join* all factors containing that variable.
 - *sum* out the influence of the variable on the new factor.

VE Example

$$\begin{aligned}P(J) &= \sum_{M,A,B,E} P(J, M, A, B, E) \\&= \sum_{M,A,B,E} P(J|A)P(M|A)P(B)P(A|B, E)P(E) \\&= \sum_A P(J|A) \sum_M P(M|A) \sum_B P(B) \sum_E P(A|B, E)P(E) \\&= \sum_A P(J|A) \sum_M P(M|A) \sum_B (P(B)f1(A, B)) \\&= \sum_A P(J|A) \sum_M P(M|A)f2(A) \\&= \sum_A P(J|A)f3(A) \\&= f4(J)\end{aligned}$$

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Q. What is the elimination order?

A. M,A,B,E

B. E,B,A,M

C. A,M,B,E

VE Example

$$P(J) = \sum_{M,A,B,E} P(J, M, A, B, E)$$

$$= \sum_{M,A,B,E} P(J|A)P(M|A)P(B)P(A|B,E)P(E)$$

$$= \sum_A P(J|A) \sum_M P(M|A) \sum_B P(B) \sum_E P(A|B,E)P(E)$$

$$= \sum_A P(J|A) \sum_M P(M|A) \sum_B (P(B)f_1(A,B))$$

$$= \sum_A P(J|A) \sum_M P(M|A)f_2(A)$$

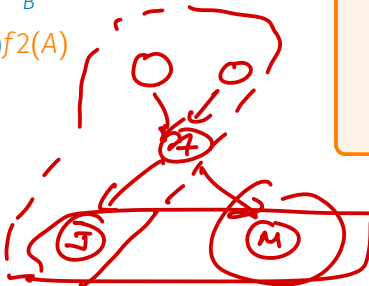
$$= \sum_A P(J|A)f_3(A)$$

$$= f_4(J)$$

$$\sum_x P(x|y) =$$

$$\sum_{M,E,B,A} P(M|A)$$

Q. What if we first eliminate M?



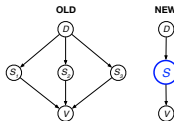
$$\sum_B P(B)$$

Exact Inference: Variable Elimination

- Time and space of VE is dominated by the **largest** factor created.
 - Can be exponential in the size of BN in the worst case.
- **Heuristic:** Eliminate the variable that will lead to the smallest next factor being created
 - In a **polytree** this leads to **linear** time inference (in size of largest CPT).

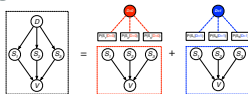
Exact Inference in Loopy BNs

- Transform a loopy BN into a polytree.
- Run the exact inference algorithm.
 - Node Clustering



CPTs grow **exponentially** when nodes are clustered.

- Cutset Conditioning



Number of runs grows **exponentially** with the size of the cutset.

- No efficient algorithm to get clustering or the minimal cutset leads to maximally efficient inference.

Approximate inference in loopy BNs

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- **Practical tools**

But many large loopy BNs remain useful models.

In these BNs, we must resort to **approximate** methods.

Motivation

- Formal results

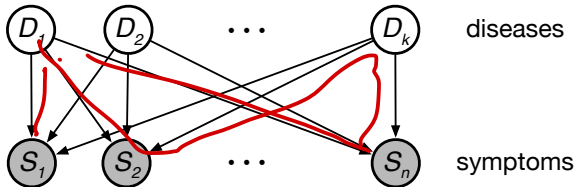
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But many large loopy BNs remain useful models.

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Approximate inference

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We will focus on **stochastic simulation methods**.



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This lecture:

1. Prior sampling
2. Rejection sampling
3. Likelihood weighting
4. Markov chain Monte Carlo (MCMC) - Gibbs Sampling

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We will focus on **stochastic simulation methods**.



This lecture:

1. Prior sampling
2. Rejection sampling
3. Likelihood weighting
4. Markov chain Monte Carlo (MCMC) - Gibbs Sampling

*But before we can describe the above methods,
we must review some basic ideas in sampling ...*

Suppose we have a (biased) coin but we don't know the $P(H)$.

- How can we estimate $P(H)$?

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- How can we estimate $P(H)$?
- When will we get the correct probability?

Basic Idea:

- Draw N samples from a sampling distribution S .

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We can apply similar intuition for inference in BNs -> Generate samples from a BN.

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Note that $P(X=x_i)$ defines a partition of unity, which maps a random number $r \in [0, 1)$ into a discrete value of X .

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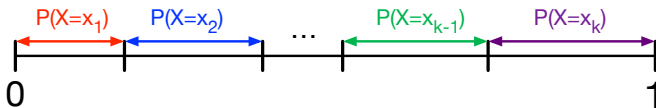
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Sampling a discrete random variable $X \sim P(X)$

C	P(C)
orange	0.6
blue	0.1
green	0.3

7

1. Get sample u from uniform distribution over $[0,1)$.
2. Assign discrete value of C.
 - $0 \leq u < 0.6, C \leftarrow \text{orange}$
 - $0.6 \leq u < 0.7, C \leftarrow \text{blue}$
 - $0.7 \leq u < 1, C \leftarrow \text{green}$

8 samples generated:

orange, green, orange, blue, orange, orange, green, orange

$$\underline{\underline{P(C = \text{green})?}}$$

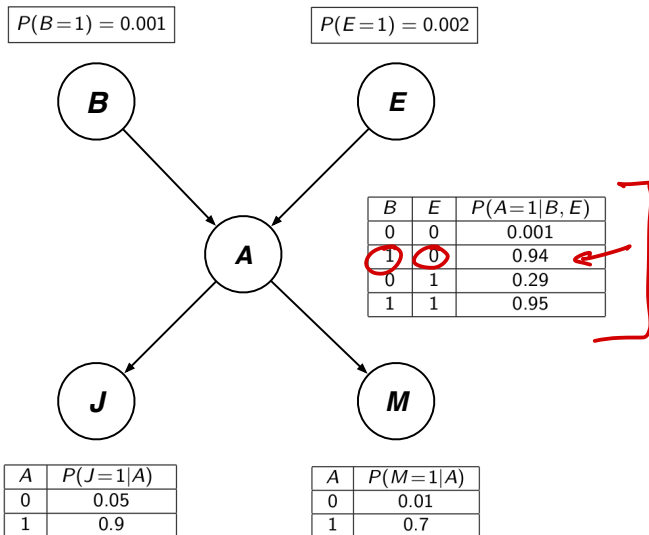
$$\frac{2}{8} = \frac{1}{4}$$

:2s

Belief Network is a **generative** model. We can sample from the distribution represented by the BN.

Idea: Generate one variable at a time in topological order.

Alarm example



Alarm example

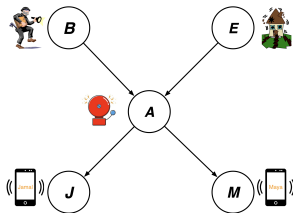
Alarm example

- Joint sample

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To draw a joint sample $\{b, e, a, j, m\}$ from $P(B, E, A, J, M)$, it is enough to draw the individual samples:

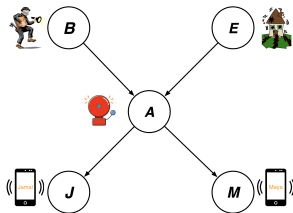


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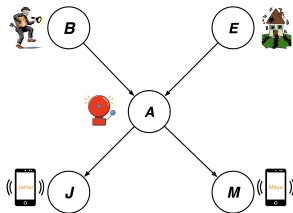
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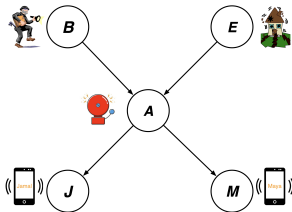
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$$a \sim P(A | \underline{B=b}, \underline{E=e}) \quad \checkmark \text{ random!}$$



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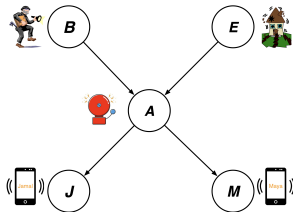
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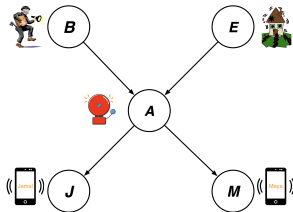
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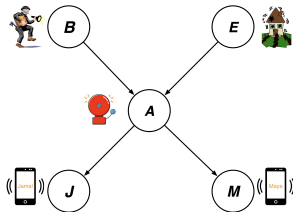
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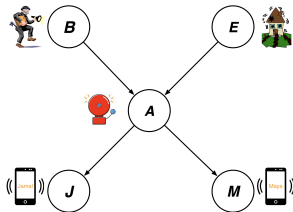
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$$P(b, e, a, j, m) =$$



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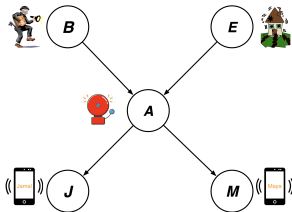
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- Convergence in the limit

$$P(b, e, a, j, m) = \lim_{N \rightarrow \infty} \frac{\text{count}(B=b, E=e, A=a, J=j, M=m)}{N}.$$



N all samples

Prior Sampling from the joint distribution in a discrete BN

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The BN defines a **generative model** with joint distribution

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To draw samples, we can simply take $X_i \sim P(X_i | \text{pa}(X_i))$.

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Approximate inference

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- **Solutions**

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- Solutions

1. rejection sampling (very inefficient)
2. likelihood weighting (more efficient)
3. MCMC (most efficient)

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Count the samples $N(q, e)$ where $Q=q$ and $E=e$.

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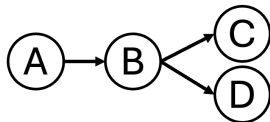
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Take the ratio of these counts:

$$\boxed{P(Q=q|E=e) \approx \frac{N(q,e)}{N(e)}} \quad \text{where} \quad \boxed{N(q,e) \leq N(e) \leq N}$$

Example for rejection sampling



A	P(A)
a_0	$1/5$
a_1	$4/5$

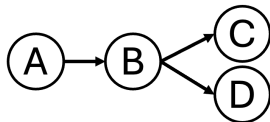
B	C	P(C B)
b_0	c_0	$1/5$
b_0	c_1	$4/5$
b_1	c_0	$3/5$
b_1	c_1	$2/5$

A	B	P(B A)
a_0	b_0	$1/4$
a_0	b_1	$3/4$
a_1	b_0	$1/3$
a_1	b_1	$2/3$

B	D	P(D B)
b_0	d_0	$3/4$
b_0	d_1	$1/4$
b_1	d_0	$1/3$
b_1	d_1	$2/3$

Example for rejection sampling

Problem: Estimate $P(a_0|c_1, d_1)$



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A	B	P(B A)
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a_0	b_1	3/4
a_1	b_0	1/3
a_1	b_1	2/3

B	C	P(C B)
b_0	c_0	1/5
b_0	c_1	4/5
b_1	c_0	3/5
b_1	c_1	2/5

B	D	P(D B)
b_0	d_0	3/4
b_0	d_1	1/4
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Example for rejection sampling

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Samples:

a_1, b_1, c_0, d_0

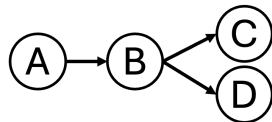
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B	C	P(C B)
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Samples:

a_1, b_1, c_0, d_0

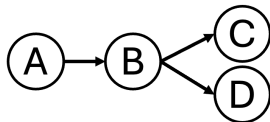
a_0, b_1, c_1, d_0

a_1, b_1, c_0, d_1

a_1, b_1, c_1, d_1

a_1, b_0, c_0, d_0

a_0, b_1, c_1, d_1



A	P(A)
a_0	1/5
a_1	4/5

A	B	P(B A)
a_0	b_0	1/4
a_0	b_1	3/4
a_1	b_0	1/3
a_1	b_1	2/3

B	C	P(C B)
b_0	c_0	1/5
b_0	c_1	4/5
b_1	c_0	3/5
b_1	c_1	2/5

B	D	P(D B)
b_0	d_0	3/4
b_0	d_1	1/4
b_1	d_0	1/3
b_1	d_1	2/3

Q. How many samples will be rejected?

A. 6

B. 2

C. 4

D. 3

E. 0

Example for rejection sampling

Problem: Estimate $P(a_0|c_1, d_1)$

Samples:

~~a_1, b_1, c_0, d_0~~

~~a_0, b_1, c_1, d_0~~

~~a_1, b_1, c_0, d_1~~

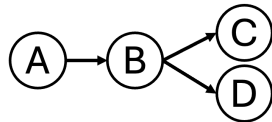
a_1, b_1, c_1, d_1

~~a_1, b_0, c_0, d_0~~

a_0, b_1, c_1, d_1

$a_1 = a_0$

2



A	P(A)
a_0	1/5
a_1	4/5

A	B	P(B A)
a_0	b_0	1/4
a_0	b_1	3/4
a_1	b_0	1/3
a_1	b_1	2/3

B	C	P(C B)
b_0	c_0	1/5
b_0	c_1	4/5
b_1	c_0	3/5
b_1	c_1	2/5

B	D	P(D B)
b_0	d_0	3/4
b_0	d_1	1/4
b_1	d_0	1/3
b_1	d_1	2/3

Q. Estimate of $P(a_0|c_1, d_1)$ using rejection sampling?

A. 1/2 B. 2/3 C. 1/4

D. 1/3 E. 0

$N(e) = 2$

$N(a_0, e) = ?$ 1

Example for rejection sampling

Example for rejection sampling

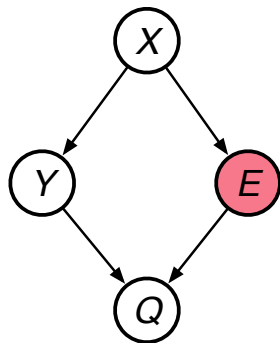
- Sample N times:

$$x_i \sim P(X)$$

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$$e_i \sim P(E|X=x_i)$$

$$q_i \sim P(Q|Y=y_i, E=e_i)$$



Example for rejection sampling

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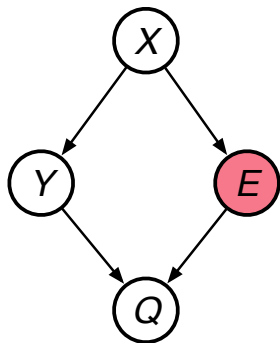
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Example for rejection sampling

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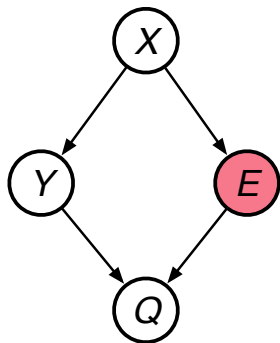
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$$I(z, z') = \begin{cases} 1 & \text{if } z = z' \\ 0 & \text{if } z \neq z' \end{cases}$$



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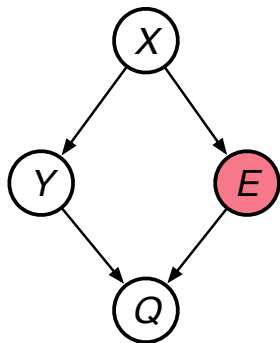
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Example for rejection sampling

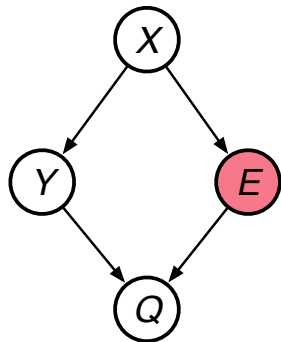
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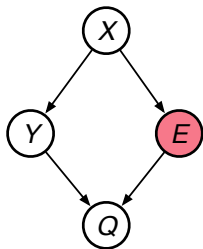
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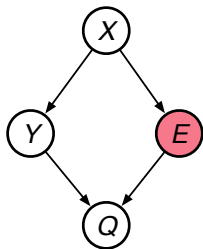
$$P(Q=q|E=e) \approx \frac{N(q, e)}{N(e)} = \frac{\sum_{i=1}^N I(q, q_i) I(e, e_i)}{\sum_{i=1}^N I(e, e_i)}$$

sample value.

Properties of rejection sampling



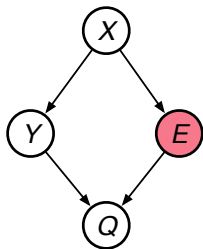
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Properties of rejection sampling

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$$\lim_{N \rightarrow \infty} \frac{N(q, e)}{N(e)} = P(Q=q|E=e)$$

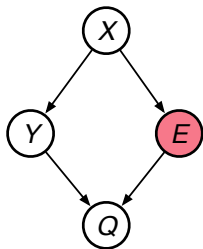


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- But it is extremely inefficient:



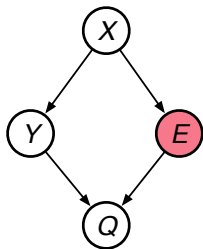
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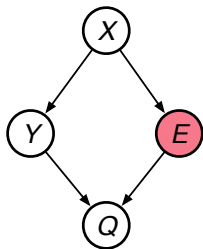
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It converges **very slowly** for rare evidence.



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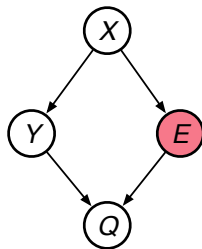
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How can we do better?

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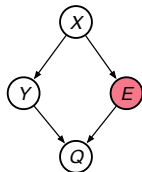
No wasted computation.

Faster convergence.

Example for likelihood weighting

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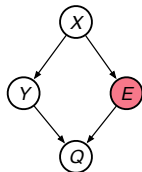
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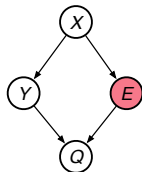


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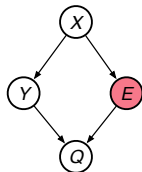
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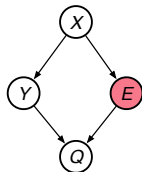
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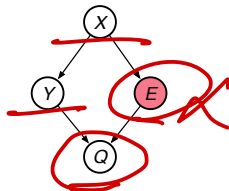
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Note: E is fixed to e .

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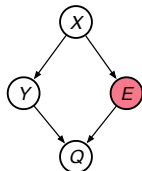
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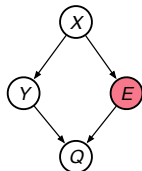
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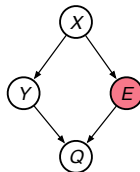
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- Estimate from ratio:

$$P(Q=q|E=e) \approx \frac{\sum_{i=1}^N I(q, q_i) \overbrace{P(E=e|X=x_i)}^{\text{likelihood weight}}}{\sum_{i=1}^N P(E=e|X=x_i)}$$

Handwritten red annotations:

- A red arrow points to the indicator function $I(q, q_i)$.
- The term $P(E=e|X=x_i)$ in the numerator is bracketed and labeled "likelihood weight" in orange.
- Handwritten red text $I(e, e_i)$ is written to the right, with a red arrow pointing to the bracketed term in the numerator.
- The denominator $\sum_{i=1}^N P(E=e|X=x_i)$ is underlined with a thick red line.

Example for likelihood weighting

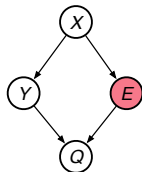
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- Compare to rejection sampling:

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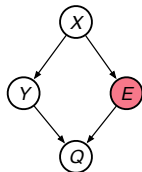
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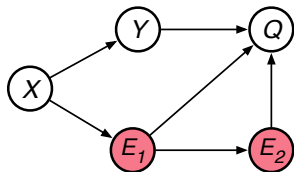
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Example with multiple evidence nodes

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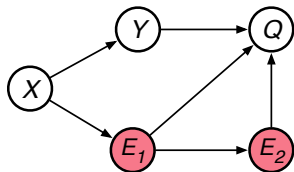
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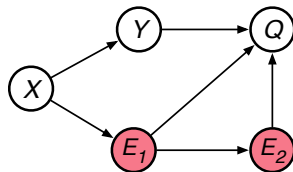


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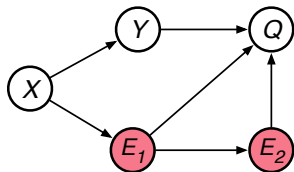
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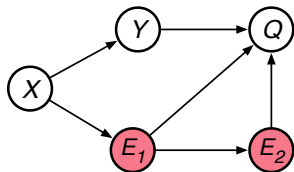
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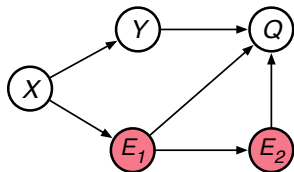
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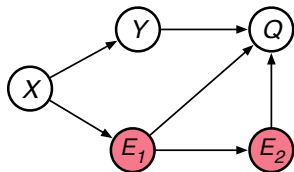
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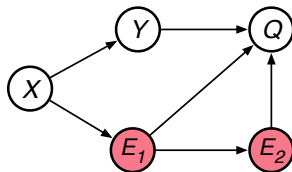
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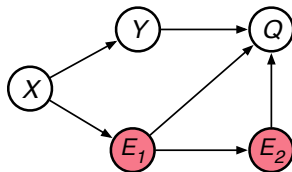
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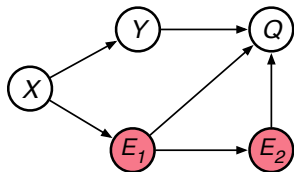
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- Compare to rejection sampling:

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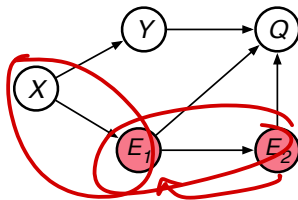
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$$P(Q=q|E_1=e, E_2=e') \approx \frac{\sum_{i=1}^N l(q, q_i) \overbrace{P(E_1=e|X=x_i) P(E_2=e'|E_1=e)}^{\text{product of likelihood weights}}}{\sum_{i=1}^N \overbrace{P(E_1=e|X=x_i) P(E_2=e'|E_1=e)}^{\text{product of likelihood weights}}}$$

- Compare to rejection sampling:

$$P(Q=q|E=e) \approx \frac{\sum_{i=1}^N l(q, q_i) l(e, e_{1i}) l(e', e_{2i})}{\sum_{i=1}^N l(e, e_{1i}) l(e', e_{2i})}$$

Example for likelihood weighting sampling

Problem: Estimate $P(a_0|c_1, d_1)$

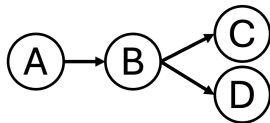
Samples:

a_0, b_1, c_1, d_1

a_1, b_0, c_1, d_1

a_0, b_1, c_1, d_1

Q. Estimate of $P(a_0|c_1, d_1)$
using likelihood weighting?



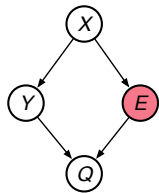
A	P(A)
a_0	1/5
a_1	4/5

A	B	P(B A)
a_0	b_0	1/4
a_0	b_1	3/4
a_1	b_0	1/3
a_1	b_1	2/3

B	C	P(C B)
b_0	c_0	1/5
b_0	c_1	4/5
b_1	c_0	3/5
b_1	c_1	2/5

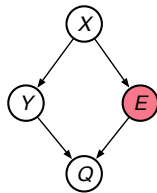
B	D	P(D B)
b_0	d_0	3/4
b_0	d_1	1/4
b_1	d_0	1/3
b_1	d_1	2/3

Properties of likelihood weighting



Properties of likelihood weighting

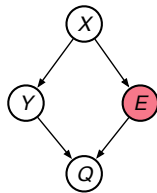
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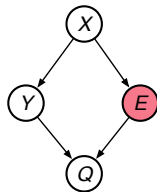
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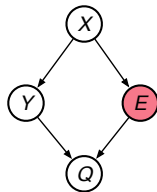
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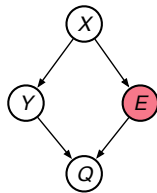


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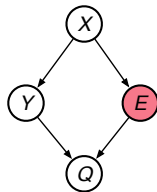
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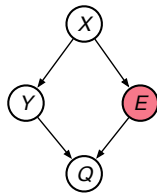
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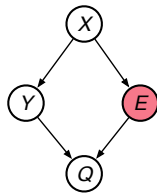
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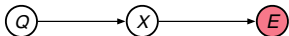


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The worst case for likelihood weighting is when rare evidence is descended from query nodes.

That's all folks!